Head Chef

CU

Control Unit

fold

You are the Head Chef Control Unit (CU)

You hold: the codebook, from page 1 of the script. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

F1 *(nothing to cook)* OK, team, let's get started. Line Counter, call out our line!

F4 *(to the Librarian)* Approved! **Read!**

D2 *(checks the codebook)* LOAD 5 means: put the amount on line five into the bowl.

E1 Caller, I need line 5.

E3 *(to the Librarian)* **Read!**

E6 *(to the Bowl)* Bowl, you are asked to hold 12 grams.

Scene 2: ADD 6

F1 OK, team, let's continue. Line Counter, call out our line!

F4 *(to the Librarian)* Approved! **Read!**

D2 *(checks the codebook)* ADD 6 means: add the amount on line six to the bowl.

E1 Caller, I need line 6.

E3 *(to the Librarian)* **Read!**

E6 *(to the Station Cook)* Station Cook, take what is in the bowl now and add the next ingredients. How much is that?

Scene 3: STORE 7

F1 OK, team, let's continue. Line Counter, call out our line!

F4 *(to the Librarian)* Approved! **Read!**

D2 *(checks the codebook)* STORE 7 means: **write** the bowl's total on line seven.

E1 Caller, please prepare the Librarian for line 7.

E3 Bowl, show the Scribe your total.

E7 *(to the Librarian)* **Write!**

Scene 4: STOP

F1 OK, team, let's continue. Line Counter, call out our line!

F4 *(to the Librarian)* Approved! **Read!**

D2 *(checks the codebook)* STOP means: the recipe is finished.

E1 *(closes the codebook)* OK, everyone, we are done! The Librarian says to stop, so we stop. Well done, team!

Line Counter

PC

Program Counter

fold

You are the Line Counter Program Counter (PC)

You hold: a mini whiteboard. Start with 0 on it. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

F2 *(to the Caller)* We're on line zero!

F7 *(in the meantime, adds 1 and writes 1, ready for the next call)*

Scene 2: ADD 6

F2 *(to the Caller)* We're on line one!

F7 *(in the meantime, adds 1 and writes 2, ready for the next call)*

Scene 3: STORE 7

F2 *(to the Caller)* We're on line two!

F7 *(in the meantime, adds 1 and writes 3, ready for the next call)*

Scene 4: STOP

F2 *(to the Caller)* We're on line three!

F7 *(in the meantime, adds 1 and writes 4, ready for the next call)*

Caller

MAR

Memory Address Register

fold

You are the Caller Memory Address Register (MAR)

You hold: a mini whiteboard, empty to start. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

F3 *(wipes the board and writes 0)*
Got it! Librarian, please prep line zero for **reading**. Await your command.

E2 *(wipes the board and writes 5)*
Got it! Librarian, please prep line 5 for reading. Await your command.

Scene 2: ADD 6

F3 *(wipes the board and writes 1)*
Got it! Librarian, please prep line one for **reading**. Await your command.

E2 *(wipes the board and writes 6)*
Got it! Librarian, please prep line 6 for reading. Await your command.

Scene 3: STORE 7

F3 *(wipes the board and writes 2)*
Got it! Librarian, please prep line two for **reading**. Await your command.

E2 *(wipes the board and writes 7)*
Librarian, please prep line 7 for **writing**. Await your command.

Scene 4: STOP

F3 *(wipes the board and writes 3)*
Got it! Librarian, please prep line three for **reading**. Await your command.

Scribe

MDR

Memory Data Register

fold

You are the Scribe Memory Data Register (MDR)

You hold: a notepad and pencil. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

F6 *(writes LOAD 5 down on the notepad)*

D1 *(holds up the notepad for the Head Chef)* LOAD 5.

E5 *(writes 12 grams on the notepad)*

Scene 2: ADD 6

F6 *(writes ADD 6 down on the notepad)*

D1 *(holds up the notepad for the Head Chef)* ADD 6.

E5 *(writes 30 grams on the notepad)*

Scene 3: STORE 7

F6 *(writes STORE 7 down on the notepad)*

D1 *(holds up the notepad for the Head Chef)* STORE 7.

E5 *(copies 42 onto the notepad)*

E6 *(walks over and hands the number to the Librarian)* Forty-two.

Scene 4: STOP

F6 *(writes STOP down on the notepad)*

D1 *(holds up the notepad for the Head Chef)* STOP.

Station Cook

ALU

Arithmetic Logic Unit

fold

You are the Station Cook Arithmetic Logic Unit (ALU)

You hold: nothing. You do the sums in your head. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

No lines in this scene.

Scene 2: ADD 6

E7 *(reads the Bowl and the Scribe's notepad)* We had 12 grams in the bowl already, sir, and when we add 30 more grams it adds up to **42 grams**. Bowl, you now have 42 grams!

Scene 3: STORE 7

No lines in this scene.

Scene 4: STOP

No lines in this scene.

Bowl

ACC

Accumulator

fold

You are the Bowl Accumulator (ACC)

You hold: the scales. They read 0 g to start. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

E7 12 grams! Got it! *(12 g is shown on the scales.)*

Scene 2: ADD 6

E8 42 grams! Got it! *(42 g is shown on the scales.)*

Scene 3: STORE 7

E4 *(shows 42 to the Scribe and keeps its number)*

Scene 4: STOP

No lines in this scene.

Librarian

RAM

Main Memory

fold

You are the Librarian Main Memory (RAM)

You hold: the recipe card. Write the total on line 7 in pencil. **Speak on your step.** Steps run F1, F2 … then D1, D2, then E1, E2 …

Scene 1: LOAD 5

F5 *(reads the line out and keeps the card)* LOAD 5.

E4 *(reads the line out and keeps the card)* 12 grams.

Scene 2: ADD 6

F5 *(reads the line out and keeps the card)* ADD 6.

E4 *(reads the line out and keeps the card)* 30 grams.

Scene 3: STORE 7

F5 *(reads the line out and keeps the card)* STORE 7.

E8 *(writes 42 in line 7's box, in pencil)*

Scene 4: STOP

F5 *(reads the line out and keeps the card)* STOP.

The Recipe Card

The Librarian holds this card. Each line's number is its address.

Line	What is written on it
0	`LOAD 5`
1	`ADD 6`
2	`STORE 7`
3	`STOP`
4	
5	`12 g`
6	`30 g`
7	`Total: _____`